# Unit ID: 679a30a4aaf593b2823f2bb5

# Unit 6 - Testing and Report

## PDF Documents

### PDF Document 1: 1738158242353-Chapter 11 - Unit 6 - Testing and Report.pdf

This PDF document is included in the unit materials.

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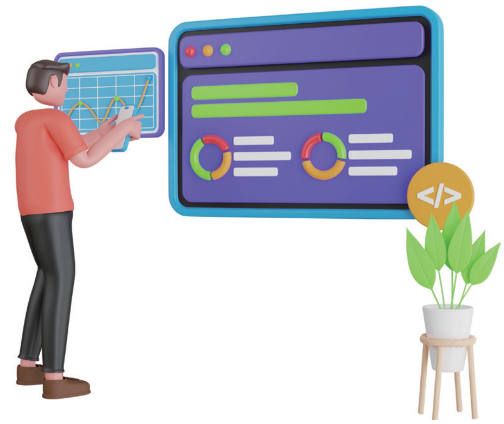
# UNIT 6 TESTING AND REPORT

## Unit Introduction

In Unit 6, you will focus on testing your smart device to ensure it functions as intended. This unit will guide you through the process of systematically testing each component and the overall system. You will also learn how to document your findings in a detailed report, highlighting any issues encountered and how they were resolved.

### WHY TESTING AND REPORTING IS IMPORTANT

Testing and reporting are very important parts of any project. Just like the devices made by companies, which go through strict testing to make sure they work correctly, your project needs to be tested too. This helps to make sure your device works as planned and is of good quality.




### LEARNING IN SCHOOL

When you go to university, testing and reporting are important in every subject. These steps help you learn how to check your work carefully, write down what you find, and make improvements. These skills are useful for doing well in school and in many jobs.

### QUALITY CHECKS IN COMPANIES

In real life, all the gadgets and devices you use, like phones and tablets, are tested a lot before they are sold. This testing makes sure they work well and are safe to use. Without this step, the devices might have problems or not work properly.

## Steps to Testing

In this testing & reporting process, you will learn how to test your smart device to make sure it works correctly. Testing is like checking your work to see if everything is right. You will go through each part of your device, see how it works, and fix any problems you find. This process helps ensure your device is reliable and functions as intended.

### Testing the Project

**1) Power on Your Device:**

- Turn on the battery module to power the system.

**2) Check Each MODI Module:**

- Verify that sensors and actuators respond as expected.
- Ensure all connections are secure and functioning.

**3) Record Initial Observations:**

- Write down your observations about how each module performs.
- Note any issues or unexpected behaviors.

**1) Stay Organized:**

- Keep your materials and notes together.

**2) Test Step-by-Step:**

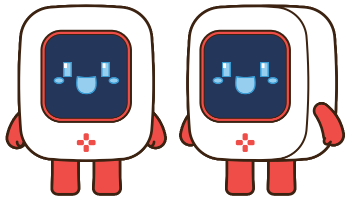
- Check each part individually.

**3) Record Everything:**

- Note your observations carefully.

**4) Repeat Tests:**

- Run multiple tests for consistency.



## EXAMPLE - OBSERVATION NOTES

**Date: Feb 21th, 2024**

**Project: Smart Plant Watering System**

**Initial Testing Observations:**

1) Power On:
   - Turned on the battery module. The device powered up without any issues.

2)Environment Module:
   - Temperature Reading: Displayed 22°C. The sensor seems to be working correctly.
   - Humidity Reading: Displayed 45%. The sensor is responding as expected.

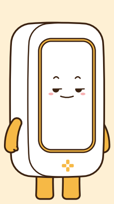
3) Display Module:
   - Display shows current temperature and humidity levels clearly.
   - No issues with the display visibility or updates.

4) LED Module:
   - LED lights up when humidity is below 30%. Tested by adjusting the sensor readings.
   - The color changes as expected to indicate low humidity.

5) Motor B (Water Pump):
   - Motor activates when humidity is below 30%. Ran the water pump for 5 seconds as intended.
   - No leaks or issues with the water delivery.

**Overall Observations:**

- Functioning Modules: All modules are functioning, but there are a few issues.
- Environment Module: The humidity reading seems inaccurate; it might need recalibration.
- Water Pump: Water delivery is inconsistent; adjustments are needed to ensure even distribution.
- Connections: The connections between modules are secure and stable.
- General Performance: No unexpected behavior or malfunctions were observed, but improvements are necessary for accurate readings and water distribution.



**Think of these observation notes as your personal project diary. Write down what you see and what happens. This way, you can look back and see how much you've learned and how far you've come. Your notes don't need to be perfect; they just need to help you remember your project's journey.**

## YOUR OBSERVATION NOTES

## Writing the Report


Reporting is a key step in completing your project. It helps you think about what you did, write down what you found, and share your results with others. A good report shows that you understand your project and have carefully thought about each part of it.

When you write a report, you're not just saying what happened—you're also explaining why it happened, what you learned, and how you can make things better. This helps you become a better problem-solver and prepares you for future projects.

In the real world, reports are used to communicate progress and findings. By learning to write a clear and detailed report, you're building skills that will be useful in school and in your future career.

**To make your report easy to follow, it's important to organize it into sections. Here's a simple structure you can use:**

## Writing the Report

### 1. Introduction:

- Start with a brief overview of your project. Explain what your device is supposed to do and why you chose to build it.
- Mention the main goals of your project.

### 2. Testing Procedures:

- Describe how you tested your device. Include the steps you followed and the tools you used.
- Explain why you chose these testing methods.

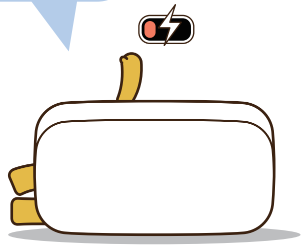
### 3. Observations and Results:

- Summarize what you observed during testing. What worked well? What didn't?v
- Include any problems you encountered and how you fixed them.

### 4. Conclusions:

- Reflect on the overall success of your project. Did it meet your goals?
- Discuss any improvements you made and what you learned from the experience.

- **Use Headings** Clearly label each section to make your report easy to read.
- **Keep It Simple** Write in short, clear sentences.
- **Use Bullet Points** This helps organize your thoughts and makes your report easier to understand.
- **Add Visuals** Diagrams or pictures can help explain your ideas better.
- **Review Your Work** Check for mistakes and make sure everything flows well.



## EXAMPLE REPORT

**Introduction:**

My project is a Smart Plant Watering System. I wanted it to water the plants when the soil is too dry.

**Testing:**

I turned on the device and checked if the sensor reads the humidity level. I also tested if the water pump starts when the soil is dry.

**Observations:**

- The sensor worked well, but the water pump didn't start.
- I found that the wire was loose, so I fixed it, and then the pump worked as expected.

**Issues and Solutions:**

- Issue: The water pump didn't start.
- Solution: I checked the connections and found a loose wire. After fixing it, the pump worked.

**What I Learned:**

- It's important to check all connections before testing.
- Fixing small issues, like a loose wire, can make a big difference in how well the project works.

**Conclusion:**

The Smart Plant Watering System worked after fixing the wire. I learned how important it is to check connections and how small fixes can improve the project's performance.

## YOUR REPORT